

```

/* =====
RaceDay Database Schema
PROG6212 POE - Part 1, Section C
Target: SQL Server (SSMS)
===== */

IF DB_ID('RaceDayDB') IS NULL
BEGIN
    CREATE DATABASE RaceDayDB;
END
GO

USE RaceDayDB;
GO

/* Drop tables if they already exist, in dependency order,
   so the script can be re-run cleanly on a clean instance. */
IF OBJECT_ID('dbo.Results', 'U') IS NOT NULL DROP TABLE dbo.Results;
IF OBJECT_ID('dbo.Enrolments', 'U') IS NOT NULL DROP TABLE dbo.Enrolments;
IF OBJECT_ID('dbo.Categories', 'U') IS NOT NULL DROP TABLE dbo.Categories;
IF OBJECT_ID('dbo.Events', 'U') IS NOT NULL DROP TABLE dbo.Events;
IF OBJECT_ID('dbo.Users', 'U') IS NOT NULL DROP TABLE dbo.Users;
IF OBJECT_ID('dbo.Roles', 'U') IS NOT NULL DROP TABLE dbo.Roles;
GO

/* =====
1. Roles
===== */
CREATE TABLE dbo.Roles (
    RoleID      INT IDENTITY(1,1) PRIMARY KEY,
    RoleName    NVARCHAR(50) NOT NULL UNIQUE
);
GO

/* =====
2. Users
===== */
CREATE TABLE dbo.Users (
    UserID      INT IDENTITY(1,1) PRIMARY KEY,
    RoleID      INT NOT NULL,
    FullName    NVARCHAR(100) NOT NULL,
    Email       NVARCHAR(150) NOT NULL UNIQUE,
    PasswordHash NVARCHAR(255) NOT NULL,
    CreatedAt   DATETIME2 NOT NULL DEFAULT SYSDATETIME(),

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        CONSTRAINT FK_Users_Roles FOREIGN KEY (RoleID)
            REFERENCES dbo.Roles(RoleID)
    );
GO

/* =====
3. Events
===== */
CREATE TABLE dbo.Events (
    EventID          INT IDENTITY(1,1) PRIMARY KEY,
    OrganiserID      INT NOT NULL,
    EventName        NVARCHAR(150) NOT NULL,
    EventDate        DATE NOT NULL,
    Location          NVARCHAR(150) NOT NULL,
    Description       NVARCHAR(1000) NULL,
    CreatedAt        DATETIME2 NOT NULL DEFAULT SYSDATETIME(),
    CONSTRAINT FK_Events_Users FOREIGN KEY (OrganiserID)
        REFERENCES dbo.Users(UserID)
);
GO

/* =====
4. Categories
===== */
CREATE TABLE dbo.Categories (
    CategoryID       INT IDENTITY(1,1) PRIMARY KEY,
    EventID          INT NOT NULL,
    CategoryName     NVARCHAR(100) NOT NULL,
    DistanceKm       DECIMAL(5,2) NOT NULL,
    MaxParticipants  INT NOT NULL DEFAULT 100,
    CONSTRAINT FK_Categories_Events FOREIGN KEY (EventID)
        REFERENCES dbo.Events(EventID)
);
GO

/* =====
5. Enrolments
===== */
CREATE TABLE dbo.Enrolments (
    EnrolmentID      INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantID    INT NOT NULL,
    CategoryID       INT NOT NULL,
    EnrolmentDate    DATETIME2 NOT NULL DEFAULT SYSDATETIME(),
    Status           NVARCHAR(20) NOT NULL DEFAULT 'Confirmed',
    CONSTRAINT FK_Enrolments_Users FOREIGN KEY (ParticipantID)
        REFERENCES dbo.Users(UserID),
    CONSTRAINT FK_Enrolments_Categories FOREIGN KEY (CategoryID)

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        REFERENCES dbo.Categories(CategoryID),
    CONSTRAINT UQ_Enrolments_ParticipantCategory UNIQUE (ParticipantID, CategoryI
);
GO

/* =====
6. Results
===== */
CREATE TABLE dbo.Results (
    ResultID          INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentID       INT NOT NULL UNIQUE,
    FinishTime        NVARCHAR(20) NULL,
    Position          INT NULL,
    Status            NVARCHAR(20) NOT NULL DEFAULT 'Finished',
    CONSTRAINT FK_Results_Enrolments FOREIGN KEY (EnrolmentID)
        REFERENCES dbo.Enrolments(EnrolmentID)
);
GO

/* =====
Seed Data
===== */

-- Roles
INSERT INTO dbo.Roles (RoleName) VALUES
    ('Organiser'),
    ('Participant');
GO

-- Users: 2 Organisers, 2 Participants
INSERT INTO dbo.Users (RoleID, FullName, Email, PasswordHash) VALUES
    (1, 'Sarah Mitchell', 'sarah.mitchell@raceday.com', 'HASHED_PASSWORD_1'),
    (1, 'David Nkosi', 'david.nkosi@raceday.com', 'HASHED_PASSWORD_2'),
    (2, 'Liam Botha', 'liam.botha@example.com', 'HASHED_PASSWORD_3'),
    (2, 'Priya Naidoo', 'priya.naidoo@example.com', 'HASHED_PASSWORD_4');
GO

-- Events: 3 events, each created by an Organiser
INSERT INTO dbo.Events (OrganiserID, EventName, EventDate, Location, Description)
    (1, 'City Spring Marathon', '2026-10-10', 'Cape Town CBD', 'Annual road
    (1, 'Riverside Fun Run', '2026-11-02', 'Riverside Park', 'Family-frie
    (2, 'Trail Blazer Challenge', '2026-11-22', 'Table Mountain', 'Off-road tr
GO

-- Categories: multiple per event
INSERT INTO dbo.Categories (EventID, CategoryName, DistanceKm, MaxParticipants) V
    (1, '10km', 10.0, 200),

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(1, 'Half Marathon', 21.1, 150),  
(1, 'Full Marathon', 42.2, 100),  
(2, '5km Fun Run', 5.0, 300),  
(2, '10km', 10.0, 150),  
(3, 'Short Trail', 15.0, 80),  
(3, 'Long Trail', 30.0, 60);
```

GO

-- Enrolments: sample participants enrolling in categories

```
INSERT INTO dbo.Enrolments (ParticipantID, CategoryID, Status) VALUES  
    (3, 1, 'Confirmed'),  
    (3, 6, 'Confirmed'),  
    (4, 2, 'Confirmed'),  
    (4, 4, 'Confirmed');
```

GO

-- Results: sample finish data for completed enrolments

```
INSERT INTO dbo.Results (EnrolmentID, FinishTime, Position, Status) VALUES  
    (1, '00:52:14', 3, 'Finished'),  
    (3, '01:58:40', 12, 'Finished');
```

GO